

Lung and Respiratory Function

Alfa-1 antitrypsin (PiZ, PiS), CHRNA5, MUC5B, FEV1, FVC, FEV1/FVC and peak flow, COPD, emphysema, pulmonary fibrosis and bronchitis, and load of lung function loci

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Genetic lung function is typical to favourable: FEV1/FVC in the percentiles 53.7 and 76.3 and FEV1 in the 42.7, with high loci load for FEV1/FVC (86.7) and FEV1 (87.2). Polygenic COPD is typical (52.3 and 43.6 percentiles; loci load in 29.5). It does not have the risk allele of MUC5B (G/G), the highest genetic factor of pulmonary fibrosis, nor the PiS of alfa-1 antitrypsin.

Unfavourable. It has alfa-1 antitrypsin PiMZ (SERPINA1 Z heterozygous; the same finding of the Liver report). MZ carriers have ~60% of normal protein; the risk of emphysema and COPD rises mainly with tobacco or tobacco. The emphysema score of 42 variants is in percentile 96.2; it probably reflects largely the Z allele itself, and the coverage is of 69%. CHRNA5 Asp398Asn is heterozygous (only relevant with tobacco). The genetic **FVC** (vital capacity) is below the median (score in percentile 28.9; loci load in the 10.7). The peak flow has mixed signals (score on 21.6, loci load on 87.2). The bronchitis is in percentile 80.4 and one of the pulmonary fibrosis scores in the 75.8 (the other in the 41.1).

What does it mean? The Sérgio F lung has typical genetics for function and a missing protection factor (PiMZ). Without tobacco or smoking, PiMZ hardly increases lung risk. Therefore, the most important rule is never smoking or vaporizing and avoiding smoke and dust. Asthma and allergies are in §47; MUC5B and AGER were also seen in §28.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 4 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 4 of the chip, 0 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 1 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 11 PGS Catalog scores calculated locally with 7.085.593 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 3082 loci, 2866 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In lung function (FEV1, FVC, FEV1/FVC, peak flow), a high percentile = better. In COPD, emphysema, pulmonary fibrosis and bronchitis, a higher percentile = more risk.

Pulmonary function

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
FEV1/FVC	PGS002692 (Weissbrod 2022, SbayesR, 985 thousand variants)	963.164 985.212 (weighted coverage 97%)	53,7	TIPIC	typical
FEV1/FVC	PGS003862 (Shrine 2023, 442 variants)	434 442 (weighted coverage 97%)	76,3	TIPIC	typical
FEV1	PGS002134 (Privé 2022, LDpred2, 940 thousand variants)	920.841 940.003 (weighted coverage 97%)	42,7	TIPIC	typical
FVC (forced vital capacity)	PGS002138 (Privé 2022, LDpred2, 940 thousand variants)	920.975 940.300 (weighted coverage 97%)	28,9	TIPIC	typical
Peak expiratory flow	PGS004434 (Jung 2024, LDpred2, 1.06 million variants)	1.036.079 1.059.939 (weighted coverage 97%)	21,6	TIPIC	typical

COPD and emphysema

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Chronic airway obstruction (COPD)	PGS002062 (Privé 2022, LDpred2, 811 thousand variants)	795.269 811.003 (weighted coverage 97%)	52,3	TIPIC	typical
DPOC	PGS001783 (Wang 2023, PRS-CS, 884 thousand variants)	859.067 884.139 (weighted coverage 97%)	43,6	TIPIC	typical
Emphysema	PGS001326 (Tanigawa 2022, 42 variants)	29 42 (weighted coverage 73%)	96,2	MAIS PROVABLE	unfavourable

Pulmonary fibrosis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Idiopathic pulmonary fibrosis	PGS005323 (Zhu 2025, PRS-CS, 653 thousand variants)	639.016 653.482 (weighted coverage 97%)	75,8	TIPIC	typical
Idiopathic pulmonary fibrosis	PGS012550 (Tang 2023, 13 variants)	12 13 (weighted coverage 68%)	41,1	TIPIC	typical

Airways

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Bronchitis	PGS019293 (Chen 2025, PRS-CS, 984 thousand variants)	950.707 983.817 (weighted coverage 96%)	80,4	MAIS PROVABLE	unfavourable

4. Key variants by topic

Variants of strong and well replicated effect on the lung.

Lung protection (antiproteases)

Alfa-1 antitrypsin protects lung tissue from neutrophil elastase. With the Z allele, the protein is retained in the liver and reaches less to the lung.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SERPINA1	rs28929474	PiZ	C/T	chip	T (1 copy/s)	unfavourable	PiZ (Glu342Lys): alfa-1 antitrypsin Z is retained in the liver and lacks in the lung. In heterozygosity (PiMZ), the risk of emphysema/COPD rises mainly with tobacco or tobacco (see also the Liver report)	Strong
SERPINA1	rs17580	PiS	T/T	chip	A (0 copy/s)	favourable	PiS (Glu264Val): slight deficiency; relevant combined with Z (PiSZ)	Strong

Smoking and nicotine

The nicotinic CHRNA5 receptor influences smoking and lung injury by smoking. Without tobacco, the effect is residual.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CHRNA5	rs16969968	D398N	G/A	chip	A (1 copy/s)	unfavourable (light)	Asp398Asn: allele A reduces the nicotinic receptor response; more tobacco consumption in smokers and more COPD and lung cancer. Without tobacco, the effect is small	Strong

Pulmonary fibrosis

MUC5B is the largest gene of idiopathic pulmonary fibrosis; its risk allele is common.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MUC5B	rs35705950		G/G	chip	T (0 copy/s)	favourable	Promoter: T allele increases the mucin MUC5B in the distal airways (the largest genetic factor of idiopathic pulmonary fibrosis, OR ~5 by allele; detail in §28)	Strong

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of lung function (FEV1, FVC, FEV1/FVC, peak flow), COPD and pulmonary fibrosis, with alleles oriented towards higher or higher risk value, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
FEV1/FVC	1107 (1028)	1098 de 2056	1020	86,7	favourable
FEV1	887 (838)	924 de 1676	831	87,2	favourable
FVC	540 (508)	512 de 1016	508	10,7	unfavourable
Peak expiratory flow	259 (241)	259 de 482	241	87,2	favourable
DPOC	209 (195)	178 de 390	195	29,5	typical
Pulmonary fibrosis / interstitial disease	80 (56)	41 de 112	56	26,5	typical

6. Favourable

- **MUC5B rs35705950 G/G**: without the risk allele of idiopathic pulmonary fibrosis (OR ~5 by allele in patients). **Strong.**
- **SERPINA1 PiS absent** (T/T): without PiSZ. **Strong.**
- **Typical to favourable pulmonary function**: FEV1/FVC in the 53.7 and 76.3 percentiles (loci load in 86.7); FEV1 in 42.7 (loci in 87.2). **Moderate.**
- **Typical COPD**: percentiles 52.3 and 43.6; loci load in 29.5. **Moderate.**

7. Unfavourable

- **SERPINA1 PiZ C/T (PiMZ)**: alfa-1 antitrypsin ~60% of normal. With tobacco, the risk of COPD rises ~1.5–2×; without tobacco, the effect is small. Strong for the genotype.
- **Emphysema**: 42 score variants in percentile 96.2 (69% coverage; probably includes the effect of Z). **Weak moderated.**
- **CHRNA5 rs16969968 G/A**: 1 allele of COPD and lung cancer in smokers. Strong for the genotype, irrelevant without tobacco.
- **Genetic FVC below the median**: score in percentile 28.9 and loci load in the 10.7 (lung volume slightly lower; normal FEV1/FVC indicates that there is no obstruction). **Moderate.**
- **Flow Peak**: score in percentile 21.6, but loci load in the 87.2 (mixed signal). **Weak.**
- **Bronchitis** in percentile 80.4; pulmonary fibrosis with mixed signs (75.8 and 41.1 percentiles; loci load in 26.5; without the MUC5B allele). **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Fragmentation of Elastin	PiMZ: less alfa-1 antitrypsin to stop neutrophil elastase in the lung	Unfavourable
Remodelling of the extracellular matrix	Emphysema (percentile 96.2 score) against fibrosis without risk MUC5B	Mixed
Protein Aggregation	Alfa-1 antitrypsin Z polymers (in the liver; see the Liver report)	Unfavourable
Chronic Inflammation	Bronchitis in percentile 80.4; Typical COPD	Unfavourable (mild)
Oxidative stress	Smoking and pollution oxidize alfa-1 antitrypsin, which aggravates PiMZ	Unfavourable (conditional)
Cell Senescence	Pulmonary fibrosis and COPD involve senescence of the alveolar epithelium; without the MUC5B allele	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Zero tobacco, e-cigarette, steam, water pipe and smoked cannabis**, and avoid passive smoking. With PiMZ is the most important rule for the lung. *Suggestion (not applied)*.
2. **Avoiding fumes and dust**: Fireplaces and firewood stoves without extraction, closed-space grilling, construction dust and solvents (FFP2/FFP3 mascara). HEPA Purifier at home; on days of poor air quality or smoke from fires, training inside and away from traffic. *Suggestion (not applied)*.
3. **Confirm the PiMZ**: alfa-1 serum antitrypsin and Pi phenotype (the same test of the Liver report). If alfa-1 antitrypsin is very low, pulmonology consultation. *Suggestion (not applied)*.
4. **Base spirometry** (FEV1, FVC, FEV1/FVC) and repeat each 3–5 years, or earlier if chronic coughing or lack of air occurs. *Suggestion (not applied)*.
5. **Updated respiratory vaccination**: annual flu, COVID-19, pertussis (dTpa every 10 years), and pneumococcal and RSV according to age. *Suggestion (not applied)*.
6. **Regular aerobic training** (zone 2 + intervals): maintains the ventilatory function and effort capacity. *Suggestion (not applied)*.
7. **Omega-3 3.2–4 g/day (plan) and vitamin D of the plan**: associated with better lung function. Fruit-rich and vegetable diet (lung antioxidants). *Suggestion (not applied)*.
8. **Respiratory infections**: to treat bronchitis early (typical-high percentile score) and not ignore the cough that lasts > 3 weeks. *Suggestion (not applied)*.

10. Limitations

- The chip reads the Z and S alleles of SERPINA1, but not the rare variants (e.g. Mmalton, null); the serum level confirms.
- The emphysema score has 69% coverage and few variants; the high percentile should be read together with PiMZ.
- The scores of pulmonary fibrosis disagree with each other; MUC5B (the greatest effect) is favourable.
- Real lung function depends mainly on exposure (tobacco, pollution, infections) and exercise.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palyndromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (11 scores: FEV1/FVC, FEV1, FVC, peak flow, COPD, emphysema, idiopathic pulmonary fibrosis and bronchitis). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** reference literature (Molloy 2014 and Strnad 2020 for PiMZ; Thorgeirsson 2008 for CHRNA5; Seibold 2011 for MUC5B; Shrine 2019 and 2023 for lung function Sakornsakolpat 2019 for COPD, GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Lung protection (antiproteases)	SERPINA1	rs28929474	C/T	chip	T	1	strand +	PiMZ: alfa-1 antitrypsin ~60% of normal; pulmonary risk mainly with tobacco/smoker
Lung protection (antiproteases)	SERPINA1	rs17580	T/T	chip	A	0	strand + (palindromic; assumed +)	without the risk allele
Smoking and nicotine	CHRNA5	rs16969968	G/A	chip	A	1	strand +	1 copy of risk allele
Pulmonary fibrosis	MUC5B	rs35705950	G/G	chip	T	0	strand +	without the risk allele